

The Complete Treatise on Solving the Linear Diophantine Equation

Lord Soumadeep Ghosh with ChatGPT (OpenAI)

Kolkata, India

Abstract

A linear Diophantine equation is a linear equation whose unknowns are required to be integers. The equation

$$ax + by = c$$

is completely characterized by the greatest common divisor of a and b . This paper develops the existence theorem, the Extended Euclidean Algorithm, Bézout's identity, modular-arithmetic methods, the complete parametrization of solutions, bounded and nonnegative solutions, and algorithmic implementations. The treatment also discusses connections with lattice geometry, integer programming, cryptography, economics, finance, physics, chemistry, biology, scheduling, statistics, computer science, and operations research. The central theorem is that integer solutions exist if and only if $\gcd(a, b)$ divides c .

The treatise ends with “The End”

Contents

1 Introduction	3
2 Problem Statement	3
3 The Fundamental Existence Theorem	3
3.1 Necessity	3
3.2 Sufficiency	4
3.3 Theorem	4
4 Bézout's Identity	4
5 The Euclidean Algorithm	4
5.1 Example	5
6 The Extended Euclidean Algorithm	5
7 A Complete Algorithm for the Diophantine Equation	6
8 The Complete Set of Integer Solutions	6
8.1 General Solution Theorem	7
9 Geometric Interpretation	7
10 Modular Arithmetic	7
11 Reduction to the Coprime Case	8
12 Special Cases	8
12.1 $a = 0$	8
12.2 $b = 0$	8
12.3 $a = b = 0$	9
13 Negative Coefficients	9

14 Nonnegative Integer Solutions	9
15 Bounded Solutions	9
16 Optimization Over the Solution Set	10
17 Economic Interpretation	10
18 Finance and Portfolio Constraints	10
19 Physics	10
20 Chemistry and Stoichiometry	11
21 Biology	11
22 Scheduling and Operations Research	11
23 Computer Science	11
23.1 Pseudocode With All Solutions in a Bounded Region	12
24 Cryptography	12
25 Statistics and Data Science	12
26 Artificial Intelligence and Machine Learning	13
27 Lattice Theory	13
28 A Deeper Algebraic Interpretation	13
29 Connection With Linear Algebra	14
30 A Worked Example From Start to Finish	14
31 Counting Solutions Under Bounds	15
32 Optimization Principle	15
33 Common Errors	15
33.1 Forgetting the Divisibility Test	15
33.2 Using Real Solutions	15
33.3 Finding One Solution and Stopping	15
33.4 Ignoring Sign or Domain Restrictions	16
34 Complexity and Algorithmic Efficiency	16
35 Generalization to More Variables	16
36 Systems of Diophantine Equations	16
37 Smith Normal Form Perspective	16
38 A Practical Decision Procedure	17
39 Conceptual Synthesis	17
40 Conclusion	18
41 A Compact Formula Sheet	18

1 Introduction

The linear Diophantine equation

$$ax + by = c,$$

where

$$a, b, c, x, y \in \mathbb{Z},$$

is one of the simplest and most important problems in elementary number theory. Its apparent similarity to an ordinary linear equation is misleading: over the real numbers there are usually infinitely many solutions, whereas requiring x and y to be integers introduces a discrete arithmetic structure.

The problem is fundamentally a question about the additive subgroup of the integers generated by a and b . That subgroup is

$$a\mathbb{Z} + b\mathbb{Z} = \gcd(a, b)\mathbb{Z}.$$

Consequently, the equation is solvable precisely when c belongs to this subgroup [1, 3, 2].

The subject connects naturally with Euclid's algorithm, Bézout's identity, modular arithmetic, lattice theory, integer programming, cryptography, resource allocation, stoichiometry, scheduling, and discrete optimization.

2 Problem Statement

Consider

$$ax + by = c.$$

We seek

$$(x, y) \in \mathbb{Z}^2.$$

Throughout most of the paper, assume that a and b are not both zero. Define

$$d = \gcd(a, b),$$

where $d > 0$.

The fundamental questions are:

1. Does an integer solution exist?
2. How can one solution be found efficiently?
3. How can all integer solutions be described?
4. What if x and y must be nonnegative?
5. What if additional bounds or optimization criteria are imposed?

3 The Fundamental Existence Theorem

3.1 Necessity

Suppose

$$ax + by = c$$

for integers x, y .

Since

$$d \mid a \quad \text{and} \quad d \mid b,$$

we have

$$d \mid ax \quad \text{and} \quad d \mid by.$$

Therefore

$$d \mid (ax + by),$$

and hence

$$\boxed{d \mid c}.$$

Thus divisibility by $\gcd(a, b)$ is necessary.

3.2 Sufficiency

Bézout's identity states that there exist integers u, v satisfying

$$au + bv = d.$$

If

$$d \mid c,$$

write

$$c = kd$$

for some $k \in \mathbb{Z}$.

Multiplication by k gives

$$a(ku) + b(kv) = kd = c.$$

Thus

$$x_0 = ku = \frac{cu}{d}, \quad y_0 = kv = \frac{cv}{d}$$

is an integer solution.

Hence:

3.3 Theorem

For integers a, b, c , with $(a, b) \neq (0, 0)$,

$$\boxed{ax + by = c \text{ has an integer solution} \iff \gcd(a, b) \mid c.}$$

This is the central solvability criterion for the problem [1, 2, 3].

4 Bézout's Identity

Bézout's identity states that for integers a and b there exist integers u and v such that

$$au + bv = \gcd(a, b).$$

The coefficients u and v are called Bézout coefficients.

This identity is more than a theoretical result: the Extended Euclidean Algorithm computes u and v efficiently.

If

$$au + bv = d,$$

and $d \mid c$, then

$$a \left(u \frac{c}{d} \right) + b \left(v \frac{c}{d} \right) = c.$$

Therefore a particular solution is

$$\boxed{x_0 = u \frac{c}{d}, \quad y_0 = v \frac{c}{d}.}$$

5 The Euclidean Algorithm

The Euclidean Algorithm computes $\gcd(a, b)$ through repeated division.

For positive integers $a > b$,

$$a = q_1 b + r_1, \quad 0 \leq r_1 < b.$$

Then

$$\gcd(a, b) = \gcd(b, r_1).$$

Repeating the process eventually produces a zero remainder. The final nonzero remainder is the greatest common divisor [4, 3].

5.1 Example

Consider

$$15x + 21y = 12.$$

The Euclidean Algorithm gives

$$21 = 1(15) + 6,$$

and

$$15 = 2(6) + 3.$$

Hence

$$\gcd(15, 21) = 3.$$

Since

$$3 \mid 12,$$

the equation is solvable.

Back-substitution gives

$$3 = 15 - 2(6),$$

and therefore

$$3 = 15 - 2(21 - 15).$$

Thus

$$3 = 3(15) - 2(21).$$

Multiplying by 4 gives

$$12 = 12(15) - 8(21).$$

Therefore one solution is

$$\boxed{(x_0, y_0) = (12, -8)}.$$

6 The Extended Euclidean Algorithm

The Extended Euclidean Algorithm simultaneously computes

$$d = \gcd(a, b)$$

and integers u, v satisfying

$$au + bv = d.$$

A recursive formulation is:

`ExtendedGCD(a,b):`

`if b = 0:`

`return (a, 1, 0)`

`(d, u1, v1) = ExtendedGCD(b, a mod b)`

`u = v1`

`v = u1 - floor(a/b) * v1`

`return (d, u, v)`

The returned values satisfy

$$au + bv = d.$$

Its running time is logarithmic in the magnitude of the input in the standard bit-complexity sense, with refinements depending on the integer arithmetic model [4, 9].

7 A Complete Algorithm for the Diophantine Equation

The following procedure solves

$$ax + by = c.$$

`SolveLinearDiophantine(a,b,c):`

```
(d,u,v) = ExtendedGCD(a,b)

if c mod d != 0:
    return "No integer solution"

k = c/d

x0 = k*u
y0 = k*v

return (x0,y0,d)
```

The output satisfies

$$ax_0 + by_0 = c.$$

All other solutions are obtained using the parametrization developed below.

8 The Complete Set of Integer Solutions

Suppose (x_0, y_0) is one integer solution:

$$ax_0 + by_0 = c.$$

Consider another solution (x, y) :

$$ax + by = c.$$

Subtracting gives

$$a(x - x_0) + b(y - y_0) = 0.$$

Hence

$$a(x - x_0) = -b(y - y_0).$$

Write

$$a = da', \quad b = db',$$

where

$$\gcd(a', b') = 1.$$

Then

$$a'(x - x_0) = -b'(y - y_0).$$

Because a' and b' are coprime,

$$b' \mid (x - x_0)$$

and

$$a' \mid (y - y_0).$$

Consequently there exists $t \in \mathbb{Z}$ such that

$$x - x_0 = b't$$

and

$$y - y_0 = -a't.$$

Thus:

8.1 General Solution Theorem

If

$$d = \gcd(a, b)$$

and (x_0, y_0) is one integer solution, then every integer solution is

$$x = x_0 + \frac{b}{d}t, \quad y = y_0 - \frac{a}{d}t, \quad t \in \mathbb{Z}.$$

Conversely, every value of $t \in \mathbb{Z}$ produces a solution.

For the example,

$$15x + 21y = 12,$$

we obtain

$$x = 12 + 7t, \quad y = -8 - 5t, \quad t \in \mathbb{Z}.$$

9 Geometric Interpretation

The real solutions of

$$ax + by = c$$

form a straight line in the Cartesian plane. The integer solutions are the lattice points lying on that line.

The direction vector of the integer solution set is

$$\mathbf{d} = \begin{pmatrix} b/d \\ -a/d \end{pmatrix}.$$

Thus

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} + t \begin{pmatrix} b/d \\ -a/d \end{pmatrix}.$$

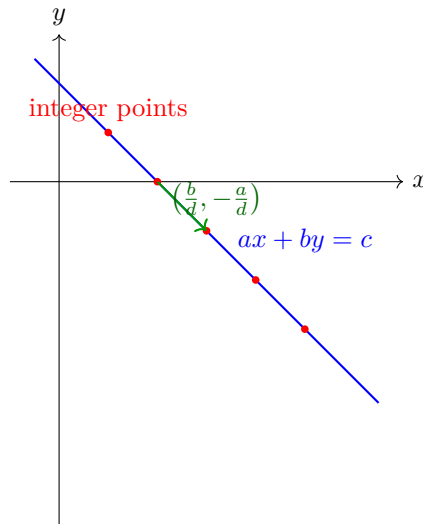


Figure 1: Geometric interpretation.

The integer points form a one-dimensional lattice embedded in the two-dimensional integer lattice $\mathbb{Z}^2$.

10 Modular Arithmetic

The equation

$$ax + by = c$$

can be reduced modulo b :

$$ax \equiv c \pmod{b}.$$

If

$$\gcd(a, b) = 1,$$

then a possesses a multiplicative inverse modulo b . Denoting this inverse by a^{-1} ,

$$x \equiv a^{-1}c \pmod{b}.$$

Therefore

$$\boxed{x \equiv a^{-1}c \pmod{b}}.$$

Similarly, reducing modulo a gives

$$by \equiv c \pmod{a}.$$

If $\gcd(a, b) = 1$, then

$$\boxed{y \equiv b^{-1}c \pmod{a}}.$$

The Extended Euclidean Algorithm computes the required modular inverse through Bézout's identity [3, 5].

11 Reduction to the Coprime Case

Let

$$d = \gcd(a, b).$$

If $d \mid c$, divide the entire equation by d :

$$\frac{a}{d}x + \frac{b}{d}y = \frac{c}{d}.$$

Set

$$a' = \frac{a}{d}, \quad b' = \frac{b}{d}, \quad c' = \frac{c}{d}.$$

Then

$$a'x + b'y = c',$$

with

$$\gcd(a', b') = 1.$$

This reduction is useful both conceptually and algorithmically.

12 Special Cases

12.1 $a = 0$

The equation becomes

$$by = c.$$

An integer solution exists precisely when

$$b \mid c.$$

Then

$$y = \frac{c}{b},$$

while x may be any integer.

12.2 $b = 0$

Similarly,

$$ax = c$$

requires

$$a \mid c.$$

Then

$$x = \frac{c}{a},$$

while y is arbitrary.

12.3 $a = b = 0$

The equation becomes

$$0 = c.$$

If $c \neq 0$, there are no solutions.

If $c = 0$, every pair

$$(x, y) \in \mathbb{Z}^2$$

is a solution.

13 Negative Coefficients

The sign of a or b does not change the underlying theory because

$$\gcd(a, b) = \gcd(|a|, |b|).$$

The same solvability condition applies:

$$\gcd(a, b) \mid c.$$

Care must be taken in implementations to use a consistent definition of the remainder operation for negative integers.

14 Nonnegative Integer Solutions

Many practical applications require

$$x \geq 0, \quad y \geq 0.$$

Starting with

$$x = x_0 + \frac{b}{d}t, \quad y = y_0 - \frac{a}{d}t,$$

the constraints become linear inequalities in t .

For positive a, b ,

$$x_0 + \frac{b}{d}t \geq 0$$

implies

$$t \geq -\frac{dx_0}{b},$$

while

$$y_0 - \frac{a}{d}t \geq 0$$

implies

$$t \leq \frac{dy_0}{a}.$$

Thus feasible integer values of t lie in an interval:

$$\left\lceil -\frac{dx_0}{b} \right\rceil \leq t \leq \left\lfloor \frac{dy_0}{a} \right\rfloor.$$

The problem of finding nonnegative solutions is therefore reduced to finding integer points in an interval.

15 Bounded Solutions

Suppose

$$L_x \leq x \leq U_x, \quad L_y \leq y \leq U_y.$$

Substituting

$$x = x_0 + \frac{b}{d}t$$

and

$$y = y_0 - \frac{a}{d}t$$

turns each bound into an inequality involving t .

The intersection of all resulting intervals gives the complete set of feasible integer values of t .

This provides an efficient method for bounded Diophantine problems without enumerating arbitrary pairs (x, y) .

16 Optimization Over the Solution Set

Suppose we wish to optimize a linear objective

$$f(x, y) = px + qy.$$

Substitution gives

$$f(t) = p \left(x_0 + \frac{b}{d}t \right) + q \left(y_0 - \frac{a}{d}t \right).$$

Therefore

$$f(t) = f_0 + \left(p \frac{b}{d} - q \frac{a}{d} \right) t.$$

Hence the objective is itself linear in t .

For a bounded feasible interval of integer t , an optimum occurs at one of the endpoints unless the objective is constant.

This is a particularly simple instance of integer linear programming [8, 7].

17 Economic Interpretation

Suppose a firm produces two products. Product A consumes a units of a scarce integer-valued resource and product B consumes b units. If exactly c units must be consumed, then

$$ax + by = c$$

models the feasible production plans, where x and y are production quantities.

The solvability condition

$$\gcd(a, b) \mid c$$

has an immediate economic interpretation: the target resource level must be compatible with the discrete increments generated by the two production technologies.

If the firm has an objective

$$\Pi = px + qy,$$

then the problem becomes

$$\max\{px + qy : ax + by = c, x, y \in \mathbb{Z}_{\geq 0}\}.$$

This is a small integer-programming model rather than a continuous linear-programming problem.

18 Finance and Portfolio Constraints

Discrete financial decisions frequently involve integer quantities, such as contracts, lots, or indivisible units. A simplified allocation constraint can have the form

$$ax + by = c, \quad x, y \in \mathbb{Z}_{\geq 0}.$$

The Diophantine condition determines whether the exact target can be reached at all.

In a more realistic model, one may impose

$$ax + by \leq c$$

and optimize a payoff or cost function. Such problems belong to integer optimization and can require methods beyond the single-equation Diophantine theory [8].

19 Physics

Quantized systems sometimes involve integer-valued conservation laws. For example, suppose two types of excitations carry conserved quantities a and b , and a total conserved quantity c is required. Then

$$ax + by = c$$

describes the admissible combinations.

The mathematical point is not that every physical system is literally a linear Diophantine equation, but that discrete conservation constraints often produce integer feasibility problems.

20 Chemistry and Stoichiometry

Chemical balancing can be formulated as a system of homogeneous linear Diophantine equations. If the stoichiometric coefficients are required to be integers, one seeks a nonzero integer vector in the nullspace of a stoichiometric matrix.

For a single conservation constraint,

$$ax + by = c,$$

the same divisibility structure appears.

More generally, if

$$A\mathbf{x} = \mathbf{0}, \quad \mathbf{x} \in \mathbb{Z}^n,$$

one encounters integer nullspaces and lattice methods. Chemical stoichiometry therefore provides a natural bridge from elementary Diophantine equations to integer linear algebra.

21 Biology

Discrete biological models can also generate equations of the form

$$ax + by = c.$$

For example, if two classes of biological units contribute fixed integer amounts to a total count, a feasibility constraint can be represented by a linear Diophantine equation.

More complicated population, genetics, metabolic, and biochemical models usually involve systems of equations or inequalities rather than a single equation. Nevertheless, the underlying question of whether a discrete conservation target can be exactly represented remains the same.

22 Scheduling and Operations Research

Suppose two activities consume a and b minutes respectively, and a schedule must consume exactly c minutes. Then

$$ax + by = c$$

is a scheduling feasibility equation.

If $a = 15$, $b = 21$, and $c = 12$, no nonnegative solution exists despite the existence of integer solutions, because every nonnegative combination is at least 15 unless both variables are zero.

This illustrates an important distinction:

Condition	Meaning
$d \mid c$	Integer solution exists
$d \mid c$ and bounds are satisfied	Feasible bounded solution exists
$d \mid c$ and $x, y \geq 0$	Nonnegative solution may exist
Optimization objective	Selects among feasible solutions

Table 1: Conditions and meanings.

23 Computer Science

The problem has several computational interpretations:

1. greatest-common-divisor computation;
2. recursive algorithms;
3. modular inversion;
4. integer programming;
5. lattice computation;
6. constraint satisfaction;
7. cryptographic arithmetic.

The Euclidean Algorithm is one of the classical examples of an efficient recursive algorithm [4, 3].

23.1 Pseudocode With All Solutions in a Bounded Region

`SolveBounded(a,b,c,Lx,Ux,Ly,Uy):`

```
(d,u,v) = ExtendedGCD(a,b)

if c mod d != 0:
    return empty set

x0 = u*(c/d)
y0 = v*(c/d)

Determine integer interval for t from:
    Lx <= x0 + (b/d)t <= Ux
    Ly <= y0 - (a/d)t <= Uy

solutions = empty set

for each integer t in the resulting interval:
    x = x0 + (b/d)t
    y = y0 - (a/d)t
    add (x,y) to solutions

return solutions
```

The algorithm avoids scanning the entire rectangle

$$[L_x, U_x] \times [L_y, U_y].$$

Instead, it reduces the two-dimensional problem to a one-dimensional integer interval.

24 Cryptography

Modular arithmetic and Bézout coefficients are central to classical public-key cryptography.

If

$$\gcd(a, m) = 1,$$

then there exist integers u, v such that

$$au + mv = 1.$$

Reducing modulo m yields

$$au \equiv 1 \pmod{m}.$$

Thus

$$u \equiv a^{-1} \pmod{m}.$$

This modular inverse operation is fundamental in many cryptographic constructions, including RSA key generation [10].

The conceptual chain is therefore

Euclidean Algorithm $\longrightarrow$ Bézout identity $\longrightarrow$ modular inverse $\longrightarrow$ cryptographic arithmetic.

25 Statistics and Data Science

Ordinary statistical models are usually formulated over the real numbers, but data-processing constraints can be discrete. Examples include integer counts, sample allocations, contingency-table margins, and resource-constrained experimental designs.

A constraint such as

$$ax + by = c$$

can represent an exact integer allocation.

In statistical computing, such constraints may occur inside integer programming, combinatorial optimization, experimental design, or sampling algorithms. The Diophantine equation is therefore best viewed as a basic building block for discrete feasibility.

26 Artificial Intelligence and Machine Learning

Many machine-learning models are continuous, but AI systems frequently interact with discrete decisions. Examples include:

- integer resource allocation;
- scheduling;
- neural-network architecture selection;
- discrete hyperparameter search;
- combinatorial optimization;
- constrained planning.

A Diophantine equation can serve as an exact discrete constraint inside a larger optimization problem. For example,

$$ax + by = c, \quad x, y \in \mathbb{Z}_{\geq 0}$$

can constrain an agent that must allocate two indivisible resources.

The single-equation case is tractable analytically; general integer optimization can be computationally much harder [8, 7].

27 Lattice Theory

The solution set has a natural lattice interpretation.

If (x_0, y_0) is one solution, then

$$S = \left\{ \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} + t \begin{pmatrix} b/d \\ -a/d \end{pmatrix} : t \in \mathbb{Z} \right\}.$$

Thus S is a one-dimensional affine lattice.

The direction vector

$$\begin{pmatrix} b/d \\ -a/d \end{pmatrix}$$

is primitive because

$$\gcd\left(\frac{b}{d}, \frac{a}{d}\right) = 1.$$

This explains why consecutive integer solutions differ by precisely this vector.

28 A Deeper Algebraic Interpretation

The set

$$a\mathbb{Z} + b\mathbb{Z}$$

is an ideal of $\mathbb{Z}$.

Every ideal of $\mathbb{Z}$ has the form

$$n\mathbb{Z}$$

for some nonnegative integer n . In this case,

$$a\mathbb{Z} + b\mathbb{Z} = d\mathbb{Z}, \quad d = \gcd(a, b).$$

Therefore

$$ax + by = c$$

is solvable exactly when

$$c \in d\mathbb{Z}.$$

The elementary divisibility theorem is consequently an instance of the ideal structure of the ring of integers [6].

29 Connection With Linear Algebra

Over the real numbers, the equation

$$ax + by = c$$

defines an affine line.

Over the integers, the relevant structure is not merely a vector space: $\mathbb{Z}$ is not a field. Consequently, division by arbitrary nonzero integers is not generally allowed while preserving integrality.

This is one of the fundamental distinctions between continuous linear algebra and integer linear algebra.

30 A Worked Example From Start to Finish

Solve

$$35x + 22y = 13.$$

First,

$$\gcd(35, 22) = 1.$$

The Euclidean Algorithm gives

$$35 = 1(22) + 13,$$

$$22 = 1(13) + 9,$$

$$13 = 1(9) + 4,$$

$$9 = 2(4) + 1.$$

Back-substitution:

$$1 = 9 - 2(4),$$

$$1 = 9 - 2(13 - 9),$$

$$1 = 3(9) - 2(13),$$

$$1 = 3(22 - 13) - 2(13),$$

and hence

$$1 = 3(22) - 5(13).$$

Since

$$13 = 35 - 22,$$

we obtain

$$1 = 3(22) - 5(35 - 22),$$

so

$$1 = -5(35) + 8(22).$$

Multiplying by 13:

$$13 = -65(35) + 104(22).$$

Thus

$$(x_0, y_0) = (-65, 104).$$

Because $\gcd(35, 22) = 1$, all solutions are

$x = -65 + 22t,$	$y = 104 - 35t,$	$t \in \mathbb{Z}.$
------------------	------------------	---------------------

For nonnegative solutions,

$$-65 + 22t \geq 0$$

requires

$$t \geq 3,$$

while

$$104 - 35t \geq 0$$

requires

$$t \leq 2.$$

There is therefore no nonnegative integer solution.

This illustrates an important distinction between:

integer feasibility $\neq$ nonnegative integer feasibility.

31 Counting Solutions Under Bounds

Suppose the admissible values of t are

$$T_{\min} \leq t \leq T_{\max}.$$

Then the number of solutions is

$$N = \max(0, T_{\max} - T_{\min} + 1).$$

Thus counting bounded solutions is reduced to computing two integer endpoints.

This reduction is an elementary example of a broader principle in discrete mathematics: exploit algebraic structure before enumeration.

32 Optimization Principle

Suppose the objective is

$$F(x, y) = px + qy.$$

Using

$$x = x_0 + \frac{b}{d}t, \quad y = y_0 - \frac{a}{d}t,$$

we obtain

$$F(t) = px_0 + qy_0 + \frac{pb - qa}{d}t.$$

Therefore:

- if $pb - qa > 0$, maximize t to maximize F ;
- if $pb - qa < 0$, minimize t to maximize F ;
- if $pb - qa = 0$, every feasible solution has the same objective value.

This is an exact analytical solution to a special class of integer-linear optimization problems.

33 Common Errors

33.1 Forgetting the Divisibility Test

It is incorrect to begin searching for solutions without first checking

$$\gcd(a, b) \mid c.$$

If this condition fails, no amount of integer searching can succeed.

33.2 Using Real Solutions

The equation always has a real solution when $(a, b) \neq (0, 0)$, but that does not imply an integer solution.

For example,

$$6x + 10y = 7$$

has infinitely many real solutions but no integer solutions because

$$\gcd(6, 10) = 2$$

and

$$2 \nmid 7.$$

33.3 Finding One Solution and Stopping

A particular solution is not the complete answer. Unless the problem explicitly asks for only one solution, the full family

$$x = x_0 + \frac{b}{d}t, \quad y = y_0 - \frac{a}{d}t$$

should be given.

33.4 Ignoring Sign or Domain Restrictions

The existence of an integer solution does not imply the existence of a nonnegative solution. Additional restrictions must be checked separately.

34 Complexity and Algorithmic Efficiency

The naive approach searches pairs (x, y) and tests

$$ax + by = c.$$

This can be extremely inefficient when $|a|$, $|b|$, or $|c|$ is large.

The Euclidean Algorithm instead reduces the problem to repeated remainders:

$$(a, b) \mapsto (b, a \bmod b).$$

The number of iterations grows logarithmically with the magnitude of the inputs under the classical arithmetic model [4].

Thus the structural algorithm is dramatically preferable to brute-force search.

35 Generalization to More Variables

The two-variable equation generalizes to

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = c.$$

Integer solutions exist if and only if

$$\gcd(a_1, a_2, \dots, a_n) \mid c.$$

For example,

$$a_1x_1 + a_2x_2 + a_3x_3 = c$$

is solvable over the integers precisely when the greatest common divisor of the coefficients divides c .

The complete parametrization now generally requires $n - 1$ independent integer parameters.

36 Systems of Diophantine Equations

A general system can be written

$$A\mathbf{x} = \mathbf{b}, \quad \mathbf{x} \in \mathbb{Z}^n.$$

This leads to integer linear algebra, lattices, Hermite normal form, Smith normal form, and integer programming [7, 6].

For a single equation in two variables, however, the entire structure is captured by the gcd and the one-parameter family derived above.

37 Smith Normal Form Perspective

For integer matrices, Smith normal form provides a canonical way of separating divisibility constraints from free integer parameters.

In the one-row case,

$$A = (a \quad b),$$

unimodular transformations can reduce the row to

$$(d \quad 0), \quad d = \gcd(a, b).$$

The equation then becomes

$$dz = c.$$

Thus solvability requires

$$d \mid c.$$

The elementary Diophantine theorem is therefore the simplest instance of a much broader canonical-form theory for integer matrices.

38 A Practical Decision Procedure

For

$$ax + by = c,$$

use the following sequence:

1. Compute

$$d = \gcd(a, b).$$

2. Check whether

$$d \mid c.$$

3. If not, stop: no integer solution exists.

4. If yes, compute Bézout coefficients u, v :

$$au + bv = d.$$

5. Construct

$$x_0 = u(c/d), \quad y_0 = v(c/d).$$

6. State the complete family:

$$x = x_0 + (b/d)t, \quad y = y_0 - (a/d)t.$$

7. If bounds or nonnegativity are imposed, convert them into inequalities for t .

8. If optimization is required, optimize the resulting linear function of t over the feasible integer interval.

39 Conceptual Synthesis

The entire theory can be summarized by the chain

Euclidean Algorithm	$\implies$	Bézout Identity	$\implies$	$\gcd(a, b)$	$\implies$	Solvability	$\implies$	Particular Solution	$\implies$	General Solution
------------------------	------------	--------------------	------------	--------------	------------	-------------	------------	------------------------	------------	---------------------

More explicitly,

$$\gcd(a, b) = d$$

determines the additive subgroup

$$a\mathbb{Z} + b\mathbb{Z} = d\mathbb{Z}.$$

Therefore

$$c \in a\mathbb{Z} + b\mathbb{Z}$$

if and only if

$$c \in d\mathbb{Z},$$

which is equivalent to

$$d \mid c.$$

Once one solution exists, every other solution differs from it by an integer multiple of

$$\begin{pmatrix} b/d \\ -a/d \end{pmatrix}.$$

Hence the problem has three logically distinct layers:

Layer	Mathematical question	Main tool
Existence	Does a solution exist?	$\gcd$ and divisibility
Construction	How do we find one?	Extended Euclidean Algorithm
Parametrization	What are all solutions?	One-parameter lattice form

40 Conclusion

The linear Diophantine equation

$$ax + by = c$$

has an exceptionally complete theory.

Let

$$d = \gcd(a, b).$$

Then:

$$ax + by = c \text{ is solvable over } \mathbb{Z} \iff d \mid c.$$

When this condition holds, the Extended Euclidean Algorithm produces integers u, v satisfying

$$au + bv = d.$$

A particular solution is

$$x_0 = \frac{cu}{d}, \quad y_0 = \frac{cv}{d}.$$

Every solution is then

$$x = x_0 + \frac{b}{d}t, \quad y = y_0 - \frac{a}{d}t, \quad t \in \mathbb{Z}.$$

This simple formula links elementary number theory to modular arithmetic, lattice geometry, algorithms, integer programming, cryptography, resource allocation, scheduling, and numerous discrete models throughout science and economics.

The deepest lesson is methodological: whenever a mathematical problem contains integer constraints, divisibility and algebraic structure should be exploited before numerical enumeration. For the two-variable linear Diophantine equation, that structure is completely captured by the greatest common divisor.

41 A Compact Formula Sheet

For

$$ax + by = c, \quad d = \gcd(a, b),$$

the essential formulas are:

Solvability:	$d \mid c.$
Bézout:	$au + bv = d.$
Particular solution:	$x_0 = \frac{cu}{d}, \quad y_0 = \frac{cv}{d}.$
General solution:	$x = x_0 + \frac{b}{d}t,$ $y = y_0 - \frac{a}{d}t, \quad t \in \mathbb{Z}.$

If

$$\gcd(a, b) = 1,$$

then

$$x = x_0 + bt, \quad y = y_0 - at.$$

References

- [1] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford University Press, Oxford, 2008.
- [2] T. M. Apostol, *Introduction to Analytic Number Theory*, Springer, New York, 1976.
- [3] K. H. Rosen, *Elementary Number Theory and Its Applications*, 6th ed., Pearson, Boston, 2011.

- [4] D. E. Knuth, *The Art of Computer Programming, Volume 2: Seminumerical Algorithms*, 3rd ed., Addison–Wesley, Reading, MA, 1997.
- [5] H. Cohen, *A Course in Computational Algebraic Number Theory*, Springer, Berlin, 1993.
- [6] D. S. Dummit and R. M. Foote, *Abstract Algebra*, 3rd ed., Wiley, Hoboken, NJ, 2004.
- [7] A. Schrijver, *Theory of Linear and Integer Programming*, Wiley, Chichester, 1986.
- [8] L. A. Wolsey, *Integer Programming*, Wiley, New York, 1998.
- [9] R. P. Brent and P. Zimmermann, *Modern Computer Arithmetic*, Cambridge University Press, Cambridge, 2010.
- [10] K. Moriarty et al., *PKCS #1: RSA Cryptography Specifications Version 2.2*, RFC 8017, Internet Engineering Task Force, 2016.

Glossary

Bézout coefficient

An integer u or v occurring in

$$au + bv = \gcd(a, b).$$

Bézout’s identity

The theorem asserting that the greatest common divisor of two integers is an integer linear combination of those integers.

Diophantine equation

An equation in which integer solutions are sought.

Extended Euclidean Algorithm

An extension of Euclid’s algorithm that computes both a gcd and Bézout coefficients.

Greatest common divisor

The largest positive integer dividing two or more integers.

Integer lattice

A discrete set generated by integer linear combinations of basis vectors.

Modular inverse

An integer u satisfying

$$au \equiv 1 \pmod{m}.$$

Particular solution

One specific solution of the Diophantine equation.

Parametrization

A representation of all solutions using one or more free integer parameters.

Unimodular matrix

An integer matrix whose determinant is ± 1 and whose inverse is also an integer matrix.

Integer linear programming

Optimization of a linear objective subject to linear constraints with some or all variables restricted to integers.

Affine lattice

A translated lattice, such as

$$\mathbf{x}_0 + t\mathbf{d}, \quad t \in \mathbb{Z}.$$

The End